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Linux Administration
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Grep

GREP is a command-line text-processing tool that allows us to search for specific patterns in a file and print the lines that contain a match. We can also use regular expressions to search for patterns instead of only exact words or characters.

Exercise 1

Task

- Print all lines containing the string Lane.

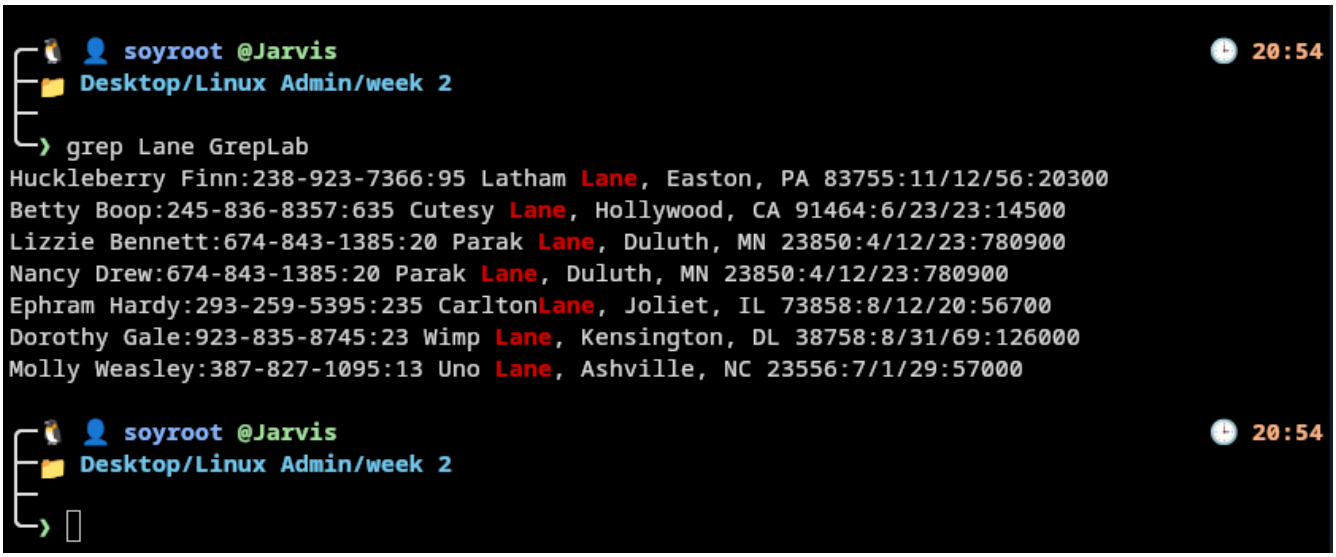
Command

- grep Lane GrepLab

How it works

- This command uses grep to search for the string Lane inside the file GrepLab. The first argument, Lane, is the pattern we want to find, and the second argument, GrepLab, is the file where we want to search. By default, grep prints every line that contains the specified pattern.

Result



Exercise 2

Task

- Print all lines where the person's first name starts with H.

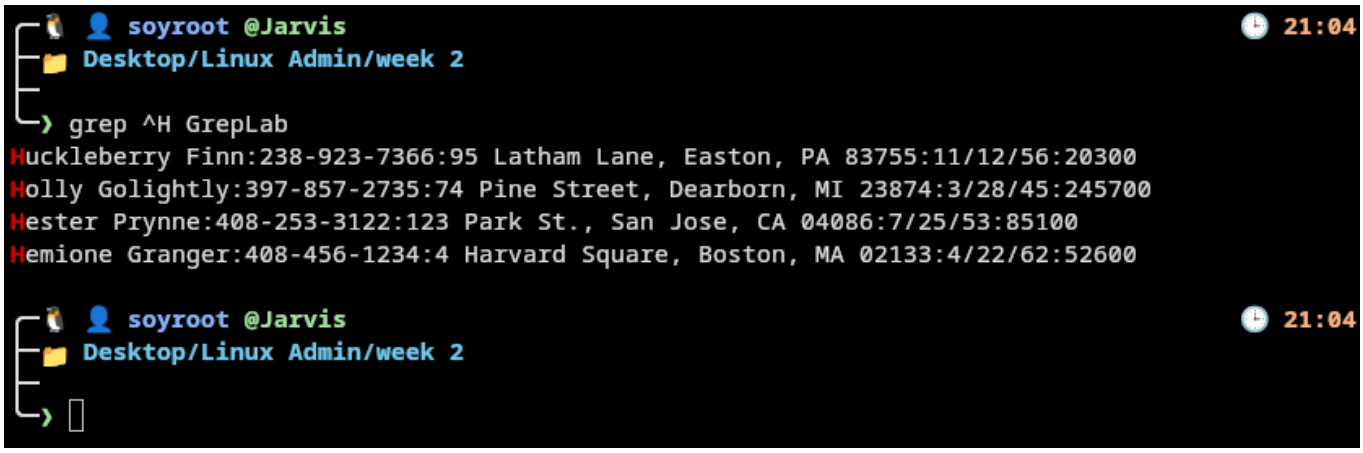
Command

- grep ^H GrepLab

How it works

- This command uses grep to search the file using the regular expression ^H. The ^ symbol represents the beginning of a line, and H is the character we want to match. This allows us to filter the lines that start with H instead of finding the letter anywhere in the line.

Result



Exercise 3

Task

- Print all lines ending with 000.

Command

- grep 000$ GrepLab

How it works

- This command uses grep to search the file for lines ending with 000. The $ symbol is a regular expression anchor that represents the end of a line. By placing it after 000, the pattern 000$ matches only lines where 000 appears at the end. GrepLab is the file where the command searches for the pattern.

Result



Exercise 4

Task

- Print all lines that do not contain 408.

Command

- grep -v 408 GrepLab

How it works

- This command uses grep with the -v option to invert the search results. Normally, grep prints all lines that contain the specified pattern, but -v makes it print the lines that do not contain it. In this case, 408 is the pattern we want to exclude, and GrepLab is the file where the command searches. The result shows all lines that do not contain 408.

Result



Exercise 5

Task

- Print all lines where birthdays are in the year 1935 (be careful of the date format! it's (MM/DD/YY)

Command

- grep /35 GrepLab

How it works

- This command uses grep to search the file for the pattern /35. The forward slash / is treated as a regular character in this search, followed by the numbers 35. Grep searches each line in the GrepLab file and prints every line containing this sequence of characters.

Result



Exercise 6

Task

- Print all lines where the phone number is in an area code that starts with an 8

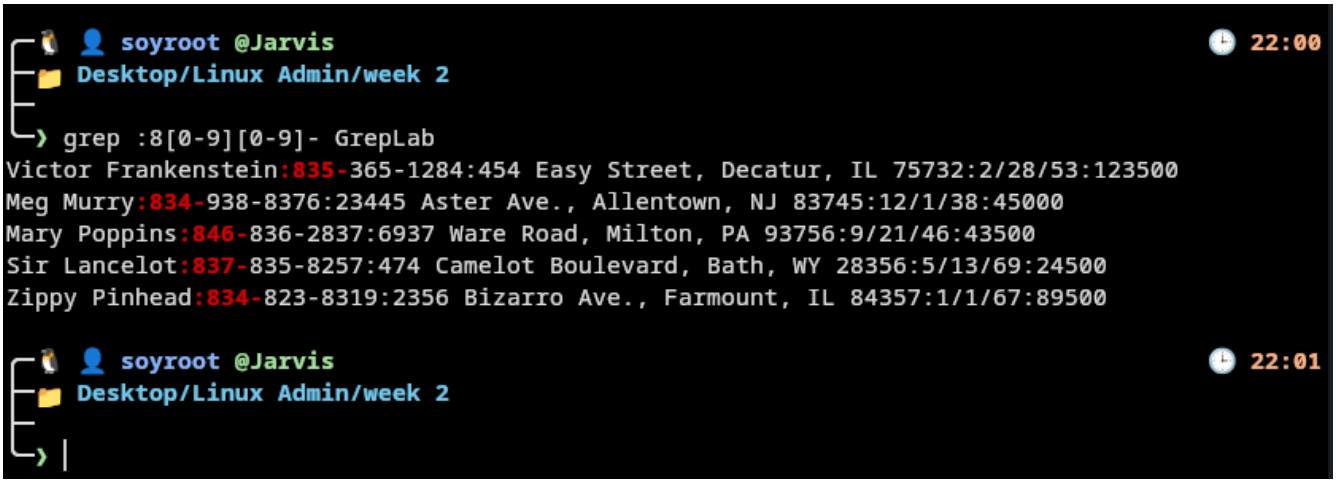
Command

- grep :8[0-9][0-9]- GrepLab

How it works

- This command uses grep to search the file for phone numbers with an area code starting with 8. The :8 identifies the beginning of the phone number in the file format. Each [0-9] is a bracket expression that matches one digit from 0 to 9, allowing any two digits after the 8. The - matches the hyphen after the area code. GrepLab is the file being searched, and grep prints all lines that match this pattern.

Result



Exercise 7

Task

- Print all lines containing an uppercase letter, followed by 4 lowercase letters, a space and one uppercase letter.

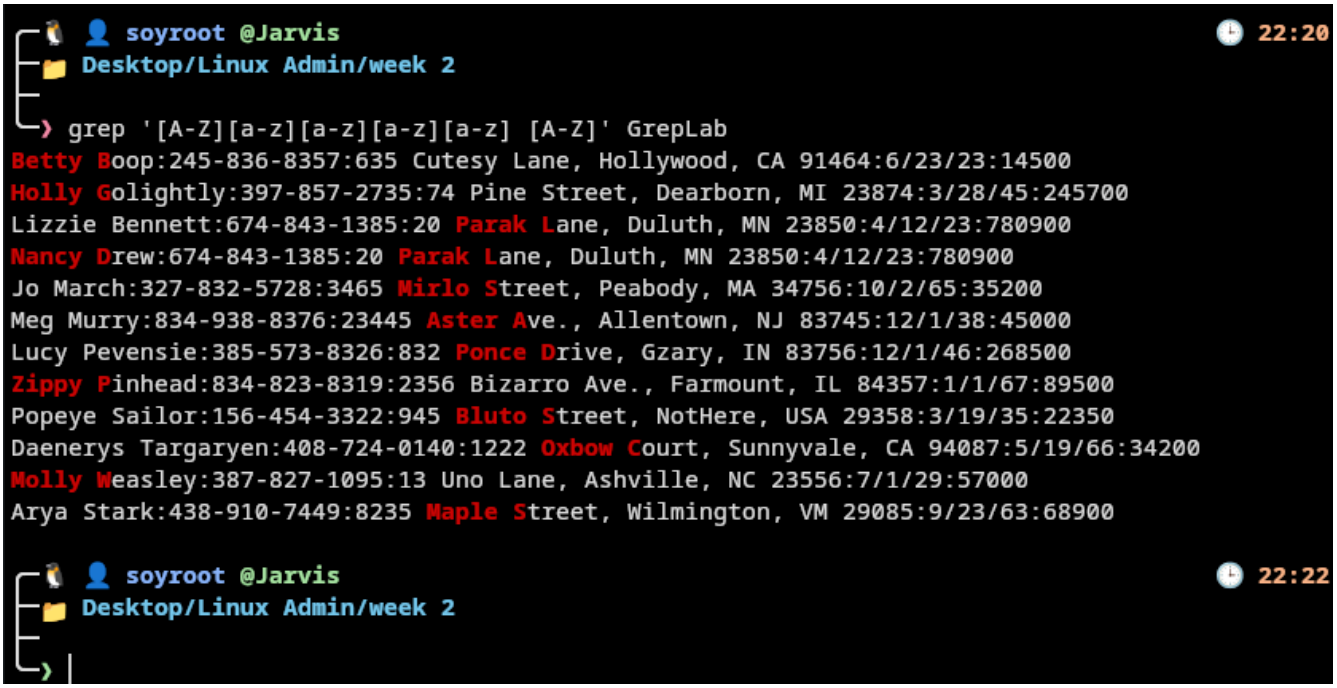
Command

- grep '[A-Z][a-z][a-z][a-z][a-z] [A-Z]' GrepLab

How it works

- This command uses grep to search the file using bracket expressions to specify the characters we want to match. [A-Z] matches one uppercase letter, followed by four [a-z] expressions, each matching one lowercase letter. The space between the expressions matches a literal space, and the final [A-Z] matches another uppercase letter. The pattern is placed in single quotes so the shell treats the entire expression, including the space, as one argument. GrepLab is the file being searched, and grep prints all lines containing this pattern.

Result



Exercise 8

Task

- Print lines where the address begins with a two or three digit number (so this would be 12 main st or 123 main street but not 1234 main street).

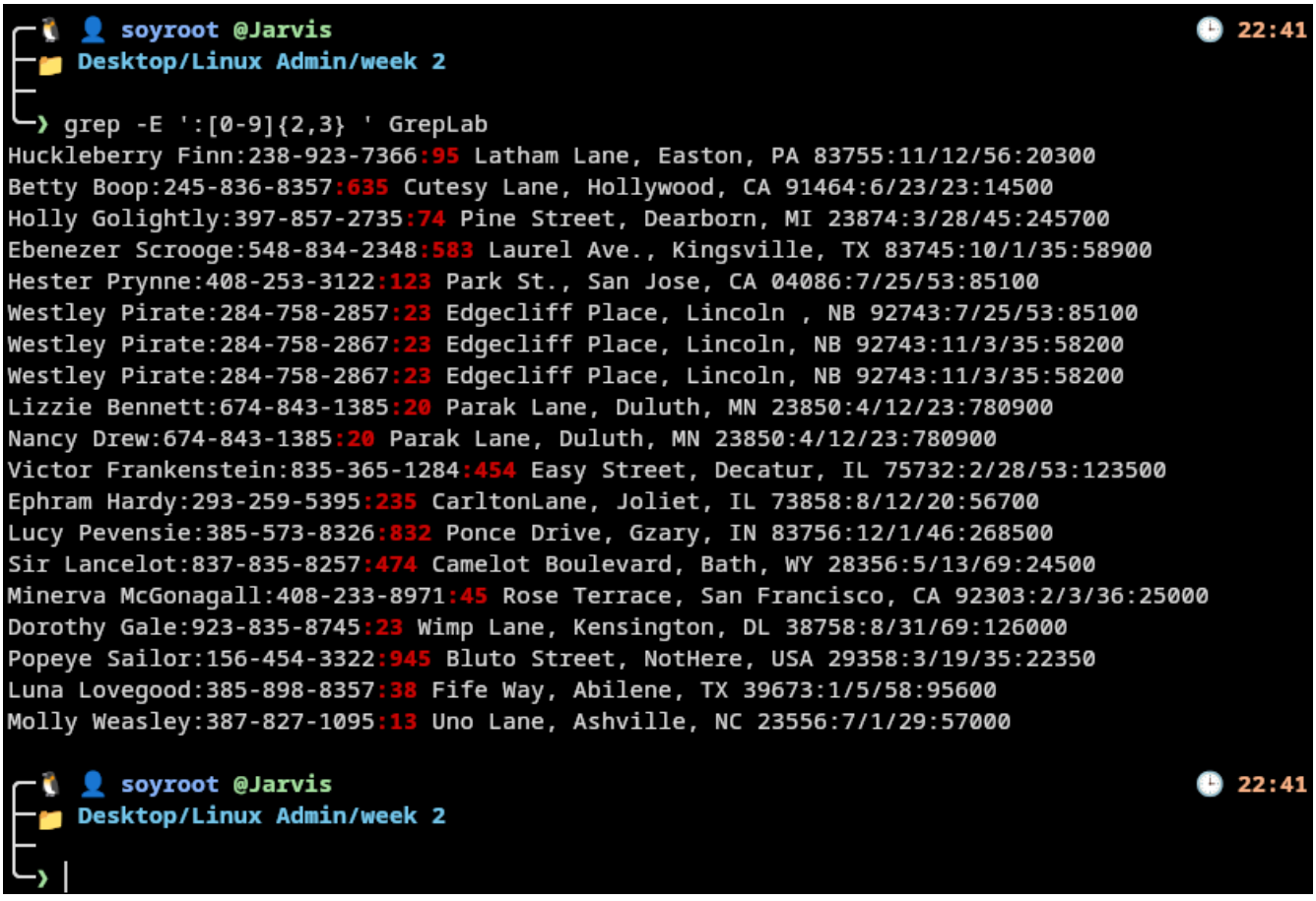
Command

- grep -E ':[0-9]{2,3} ' GrepLab

How it works

- This command uses grep with the -E option to enable extended regular expressions. The colon : identifies where the address begins in the file. The bracket expression [0-9] matches any digit from 0 to 9, and {2,3} specifies that the digit must appear two or three times. The space after the expression makes sure the number is followed by a space, excluding addresses that begin with four-digit numbers. The pattern is placed in single quotes so the shell treats it as one argument. GrepLab is the file being searched, and grep prints the lines that match the pattern.

Result



Exercise 9

Task

- Print lines where the person lives in Mass or Illinois

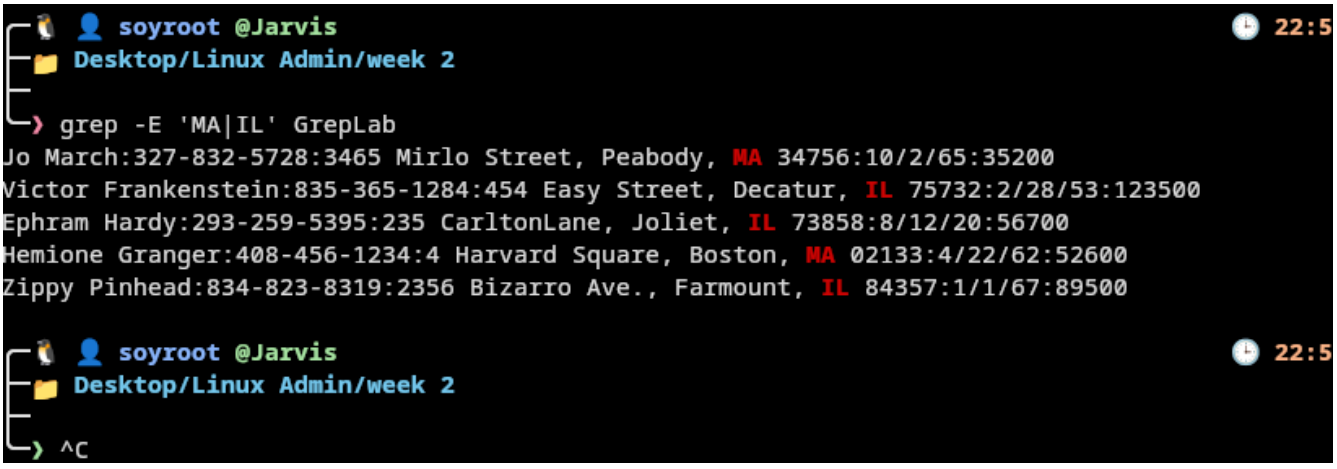
Command

- grep -E 'MA|IL' GrepLab

How it works

- This command uses grep with the -E option to enable extended regular expressions. The pattern MA|IL uses the | symbol as the OR operator, allowing grep to match either MA or IL. The pattern is placed in single quotes so the shell treats the entire expression as one argument. GrepLab is the file being searched, and grep prints all lines containing either of these patterns.

Result



Exercise 10

Task

- Print lines containing the addresses that aren’t on a street (You might see St as shorthand for street)

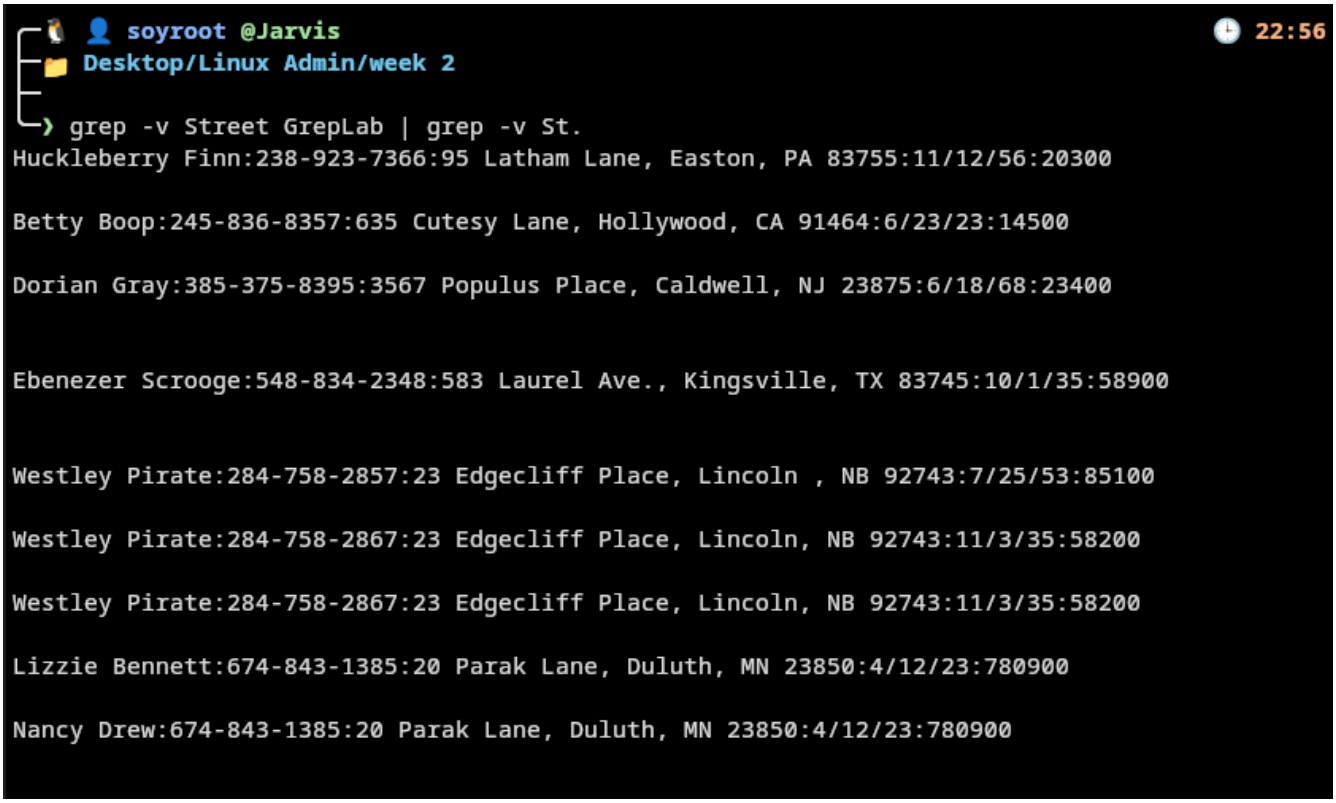
Command

- grep -v Street GrepLab | grep -v St.

How it works

- This command uses grep with the -v option to exclude lines containing the pattern Street. The pipe | takes the output from the first command and passes it to a second grep command. The second grep uses -v again to exclude lines containing the pattern St., leaving only the lines that do not match either pattern. GrepLab is the file being searched by the first command.

Result



Sources

Holden-Gouveia, Adrianna. Grep. Linux Administration.

https://www.aholdengouveia.name/LinuxAdmin/Grep.html

GNU Project. GNU Grep Manual. Free Software Foundation.

https://www.gnu.org/software/grep/manual/grep.pdf

Goyvaerts, Jan. Regular-Expressions.info: Regex Tutorial.

https://www.regular-expressions.info/quickstart.html